

TODS BRIEFING

Connect Your Scheduling and Dispatch Systems

July 14, 2026

MobilityData is a global non-profit dedicated to advancing open transportation data standards.

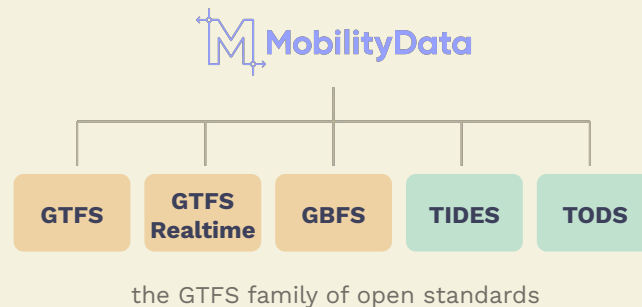


- We develop community-driven open data formats such as [GTFS](#) and [GBFS](#), and offer free tools to improve data accessibility, quality, and reuse.
- Our goal is to empower sustainable mobility actors to make a better use of their most precious resource: data.



MobilityData, the home of open transit data standards

- **MobilityData:** international non-profit, steward of the GTFS family of open standards (GTFS, GTFS Realtime, GBFS, and more)
- TODS program management since 2024
- **Membership:** how organizations support and shape this work



Today's speakers



Josh Fabian

Deputy Director of Transit
Technology, Data
MBTA



Jeff Kessler

Director of Innovation
Keolis Commuter Services



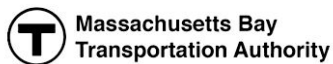
Timon Kelter

R&D Team Lead
Optibus



Anne-Sophie Lachapelle

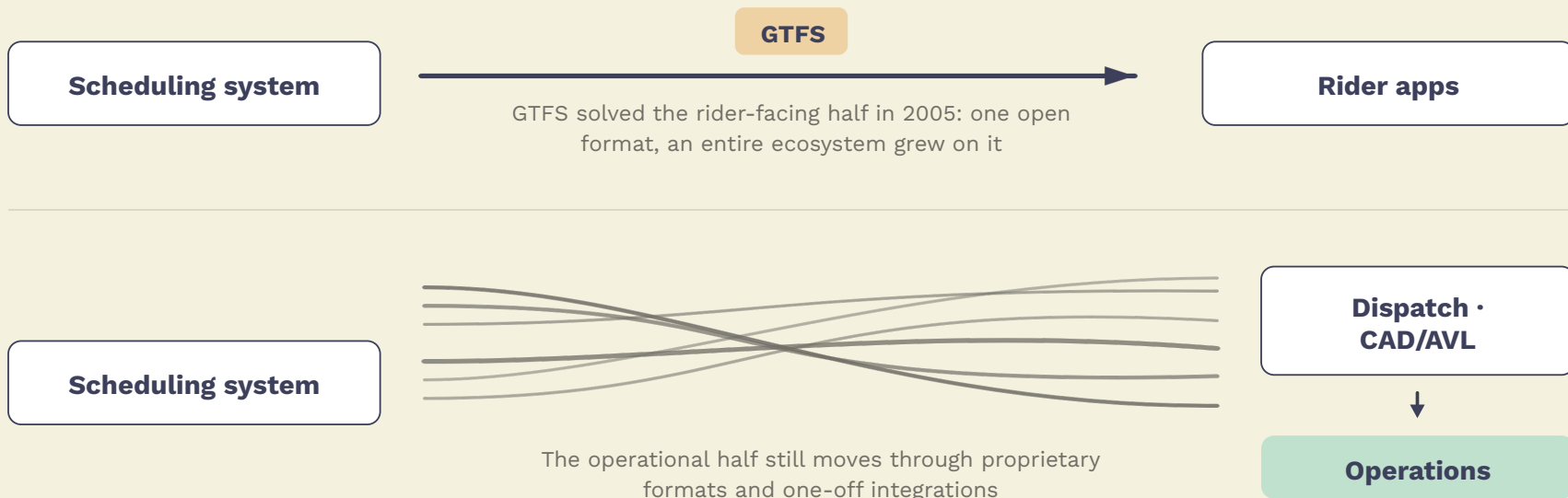
Master's researcher
Polytechnique Montréal



Facilitated by **Carl Fredlund**, MobilityData



The systems that run transit don't share a language



TODS is the open standard for how scheduled service is operated

- **TODS** = Transit Operational Data Standard
- Extends the GTFS an agency already produces
- **An open standard, not open data:** the specification is public and free to implement; the data you produce with it stays internal, unlike your published GTFS

Supplement files add to **and modify** the public GTFS with the internal picture

trips_supplement.txt

stops_supplement.txt

stop_times_supplement.txt

calendar supplements

Optional TODS-specific files carry what GTFS has no place for · include if desired

run_events.txt

employee_run_dates.txt

vehicles.txt

vehicle_assignments.txt

crew runs · vehicles and assignments

GTFS the public feed you already produce

GTFS + (and modify) TODS = Truth



Public information

GTFS: trips, routes, schedules

Internal information

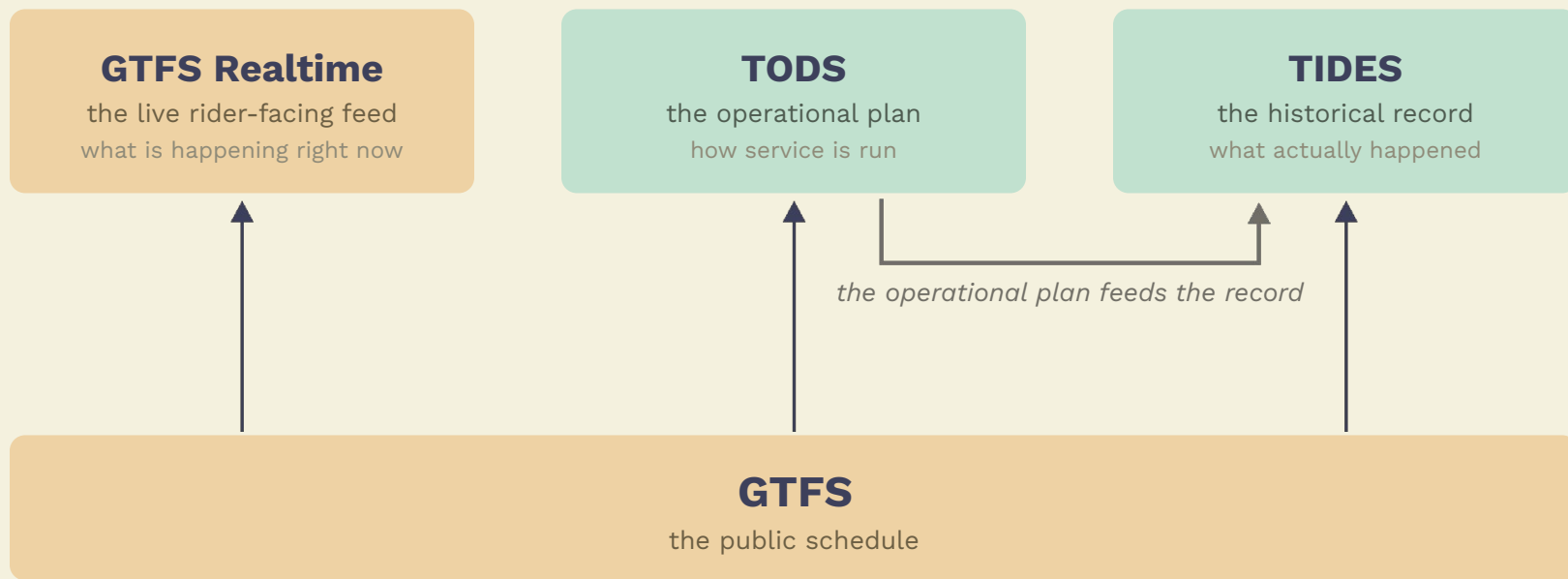
Adds: deadheads, crew, vehicles, non-public stops

Modifies: internal times, names, identifiers

The full operational picture

after Jeff Kessler, “The Journey of TODS”

Built on GTFS: the standards family

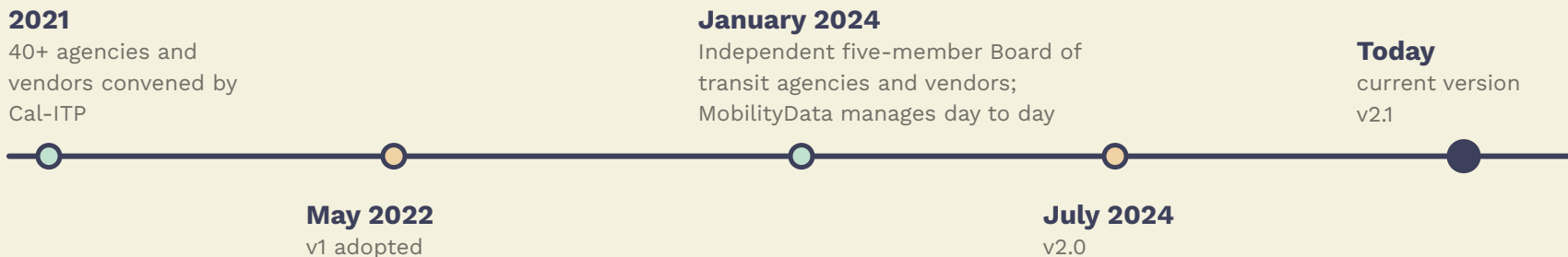


TODS and TIDES stewarded by MobilityData



Community-built, independently governed, in use

Origins at the California Integrated Travel Project (Cal-ITP)



In use and in progress across agencies and the scheduling and CAD/AVL vendor space · per tods-transit.org



Josh Fabian

Deputy Director of Transit Technology, Data
MBTA

The large-agency journey with TODS at MBTA

The MBTA, and where operational data lives

Subway

Bus

Commuter Rail

Ferry

Paratransit

- One of the largest transit agencies in the US
- Commuter Rail operated by Keolis Commuter Services
- **Transformation & Technology:** department where scheduling, dispatch, and data systems meet



February 2026: a blizzard, an emergency schedule, and systems out of sync

Scheduling $\neq$ Dispatch

at 4:00 AM

- Emergency snow schedules activated across systems the afternoon prior
- A schedule mismatch between systems took down **Skate**, our bus operations tool, for the first four hours of service
- The lesson: when operational data has no shared language, the failure shows up at the worst moment

An honest status: partial, real, and moving

- Partial implementations of TODS v1 and v2 in place; not yet complete, not yet full use
- Working with our internal scheduling support team to grow coverage
- MBTA **aims to put TODS to work** across how the agency plans and runs service
- The point for other agencies: you don't have to be finished to start getting value

Spring 2022

Began
implementing
TODS v1

Summer 2022

Started flagging possible
limitations of the
specification

2023–2024

Collaborated on
building TODS v2
specification

Today

Working version of
TODS v2 supplement
files

Now–2027

Full build-out of TODS,
integration into
first-party dispatch apps

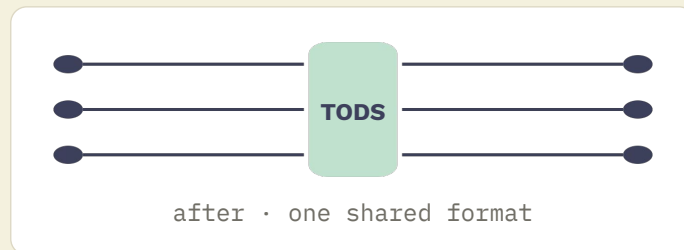
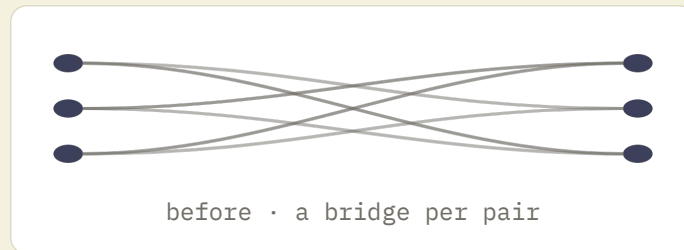
2027+

Third-party
vendor integration
of TODS



What changes when the operational layer is standard

- Systems stay in sync because they speak one format, not N different integrations
- Your operational data becomes portable, like GTFS; switching a vendor stops being a re-platforming project
- **Procurement leverage:** ask for TODS output/input when you buy or renew scheduling or CAD/AVL package
- Start with a focused slice; the standard grows with use





Massachusetts Bay
Transportation Authority

Josh Fabian

Deputy Director of Transit Technology, Data
MBTA Transformation & Technology Department

TODS Board of Directors Coordinator

jfabian@mbta.com
[linkedin.com/in/jfabi](https://www.linkedin.com/in/jfabi)



Jeff Kessler

Director of Transit Innovation

Keolis Commuter Services

From Unmet Need to New Data Standard: **The Journey of TODS**

From Unmet Need to New Data Standard: The Journey of TODS

Jeff Kessler, Director of Innovation @
Keolis Commuter Services (MBTA CR Operator)
TRB AP017 Member & TODS Board Member

July 2026





Transit systems rely on technical systems talking together

GTFS standardized information exchange for the needs of riders; TODS does this for internal operations

Vehicle & Crew Schedules

GTFS doesn't reflect the full operation. What's Missing?

- Internal "Deadhead" trips
- Crew information
- Equipment information

GTFS

Custom

Custom

Custom

Custom

Custom

Clocks

Signs

Mobile Apps

Website

Ticket Sales

Incident Response

Onboard Info

GPS Tracking

OTP

Dispatch

Simulation

Crew Info

Fuel Use

Engine Idling



But even for public trips, public data doesn't tell the whole story

Even "public" trips require "non-public" data, particularly for Railroads

Public trips making
non-public stops
(e.g. train yards)

Passing times at
interlockings
(switches)

**Different public vs
internal times** for
meets

Public routes
subdivided into
multiple internal routes

Terminal "**block**" &
track information that
regularly varies

Internal place names
that differ from public
names



But even for public trips, public data doesn't tell the whole story

Even "public" trips require "non-public" data, particularly for Railroads

Public trips making
non-public stops
(e.g. train yards)

Passing times at
interlockings (switches)

**Different public vs
internal times** for meets

Public routes subdivided
into multiple internal routes

Terminal "**block**" & **track
information** that regularly
varies

Internal place names that
differ from public names

v2's supplement files add to **and modify** the public GTFS

GTFS row

+

supplement row

=

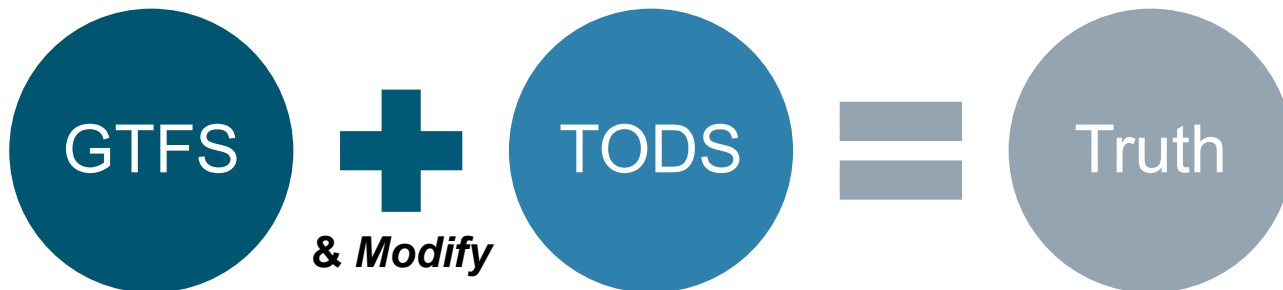
merged operational row

that is the "(and modify)" in the equation



"TODS" v2+ solves the key problem

TODS allows users to add and modify their GTFS data with relevant non-public information critical to their operation



Public Information

Trips & Schedules
Routes
Fare Information

Internal Information

Deadhead Trips
Non-Public Stops
Internal References
Crew Runs
Vehicle & Employee Data

Accurate Model

Fully-Accurate Trips
System-Level View
Context-Aware Terms



Massachusetts Bay
Transportation Authority

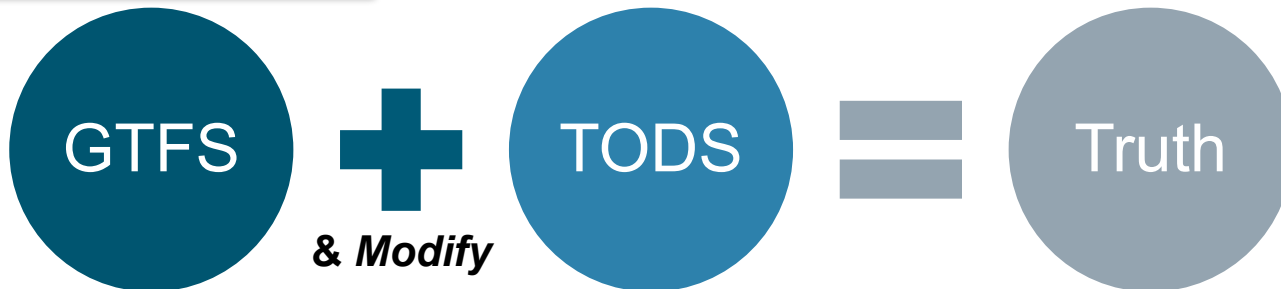
KEOLIS



An easy standard to implement

Start with existing GTFS, add supplements as required, then modify trips as needed

1. Your Already Produce!



Public Information

Trips & Schedules
Routes
Fare Information

Internal Information

Deadhead Trips
Non-Public Stops
Internal References

Crew Runs
Vehicle & Employee Data

2. Layer the Supplements

trips_supplement.txt | stop_times.txt
stops_supplement.txt

3. Add Crew/Vehicle Files as Needed

run_events.txt | vehicles.txt
vehicle_assignments.txt



Massachusetts Bay
Transportation Authority

KEOLIS



Open Standards ≠ Open Data

TODS is an Open Standard designed for Internal (Non-Public) Data

Open Standard

Data is **Structured and Exchanged** in a freely-available, unified format

Open Data

Underlying Data is freely available for public use

Can benefit from network effects of a data standard without needing to publicly release internal data



Why should I care?

Adopting TODS provides benefits on all sides

Agency Executives

Fund Tools, Not Integration: Dollars go towards actual products, not re-creating data exchanges.

Avoids Vendor Lock-In: easier & less risky changes when replacement systems already speak the same language.

Economies of Scale: TODS-based developments made for other agencies can be easily replicated at your agency.

Industry Vendors

Easier RFP Response: Lower barrier-to-entry when less customization needed to integrate.

Greater Predictability: Working with data standards means fewer unknowns when integrating a system.

Simpler Implementations: “Plug-and-play” without needing to design custom data exchanges.

Cost Amortization: Feature enhancement costs can be diversified amongst all TODS-using customers, rather than needed solely for a single customer.



Let's talk TODS

Have questions? Feel free to reach out!



+ Jeff Kessler

- > [linkedin.com/in/jeffkess](https://www.linkedin.com/in/jeffkess)
- > jeffrey.kessler@keoliscs.com

*Member of TRB AP017: Transit Data and Emerging Technologies Committee
& Transit Operational Data Standard (TODS) Board of Directors*



Massachusetts Bay
Transportation Authority

KEOLIS



Timon Kelter

R&D Team Lead

Optibus

Connecting to/from Optibus products via TODS



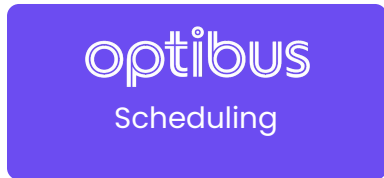
Connecting to/from Optibus products via TODS

Timon Kelter, R&D Team Lead, Optibus



Use case 1

Connect Optibus scheduling to AVL system



AVL system

GTFS base data

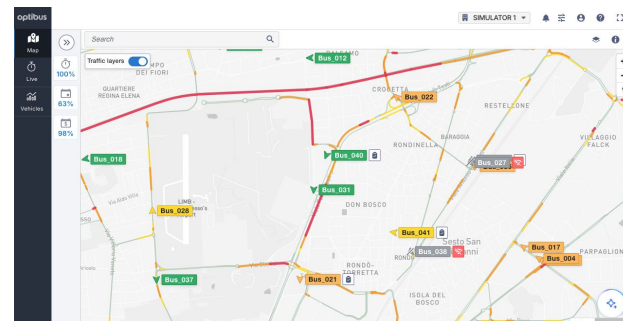
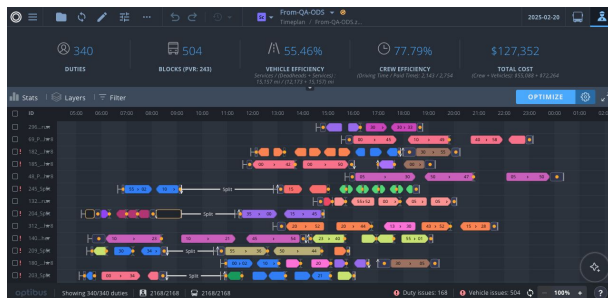
run_events.txt

trips_supplement.txt

stop_times_supplement.txt

routes_supplement.txt

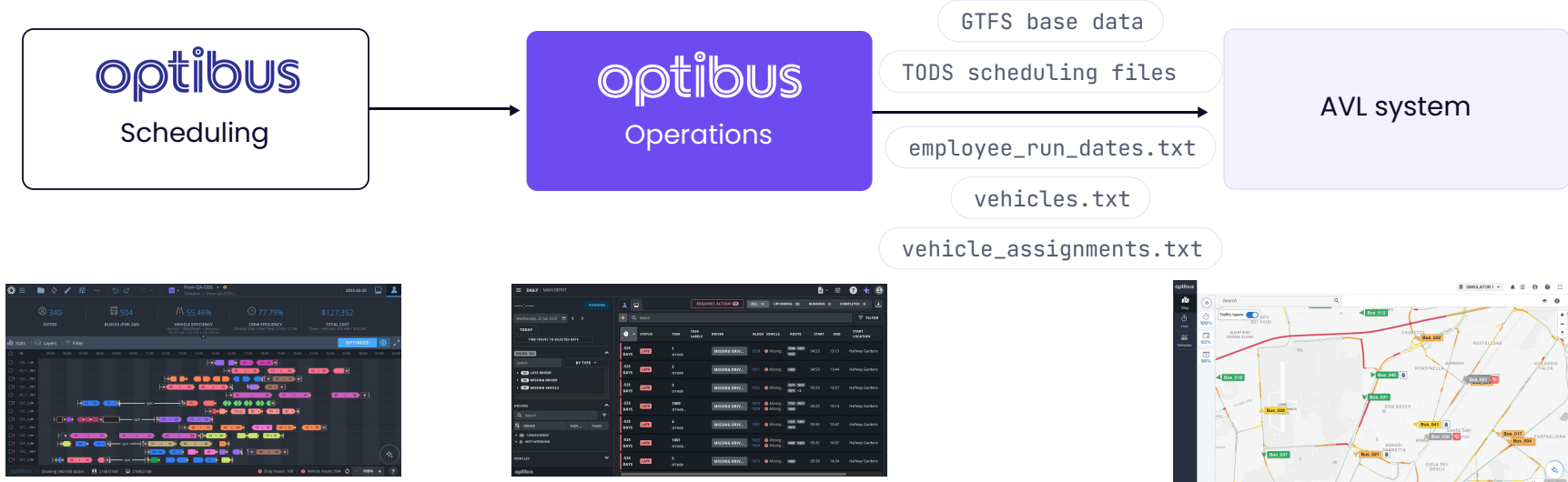
stops_supplement.txt



- **Content:** Network, routes and basic operational plan
- **Example use case:** Setup new timetable period

Use case 2

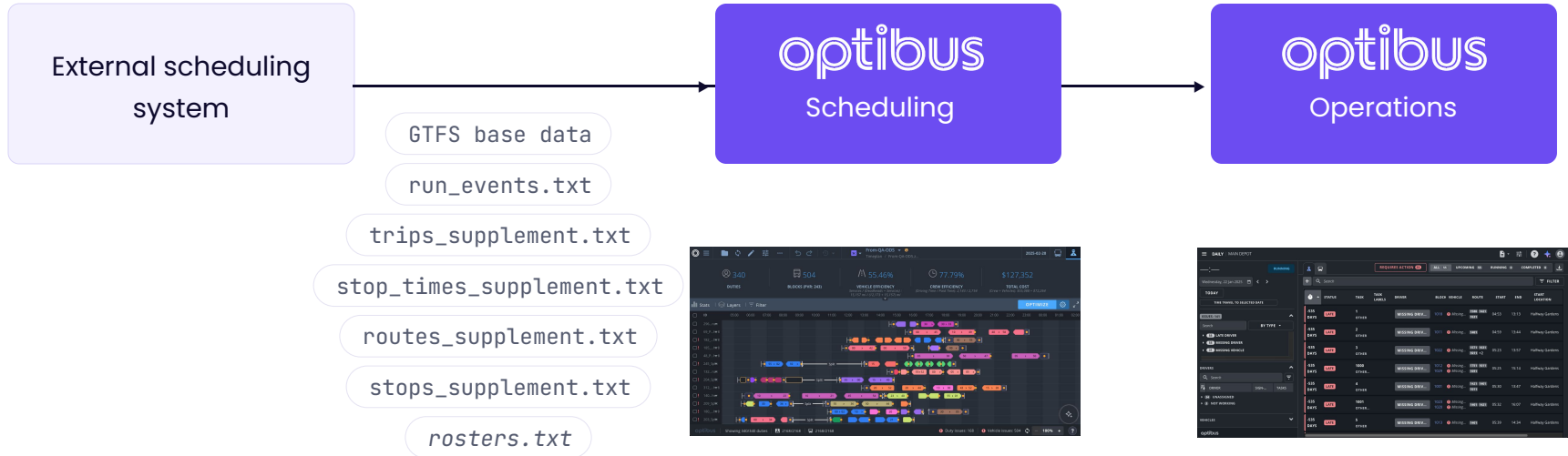
Connect Optibus operations to AVL solutions



- **Content:** Full operational plan (incl. assignments)
- **Example use case:** Copy operational plan to AVL for X next days

Use case 3

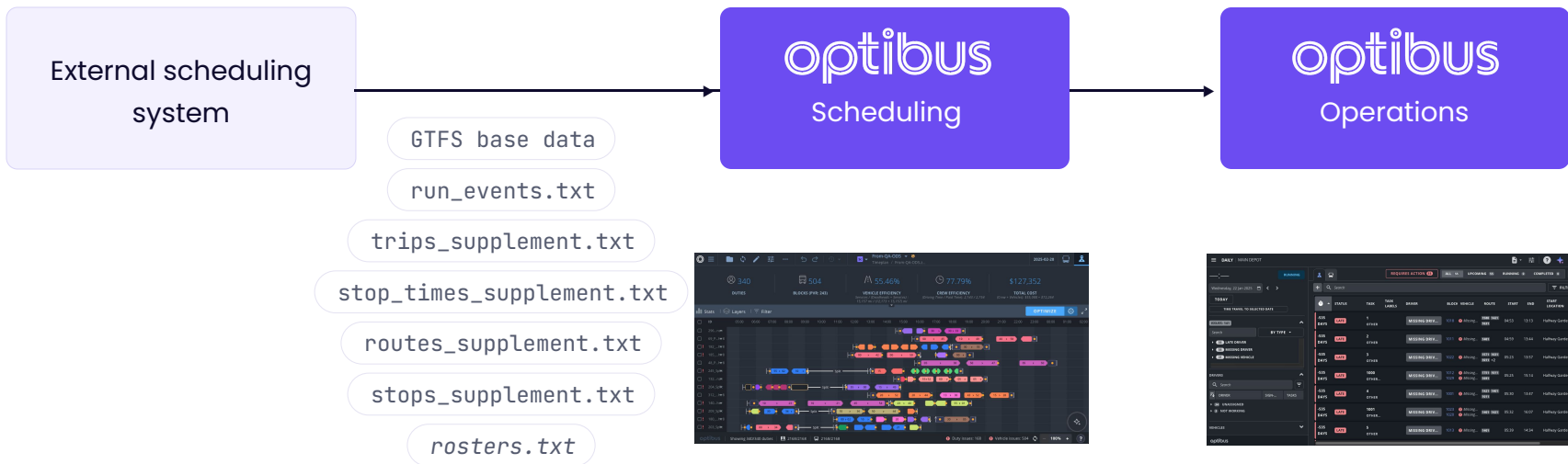
Import operational data into Optibus scheduling



- Work with existing schedules in Optibus Ops

Use case 3

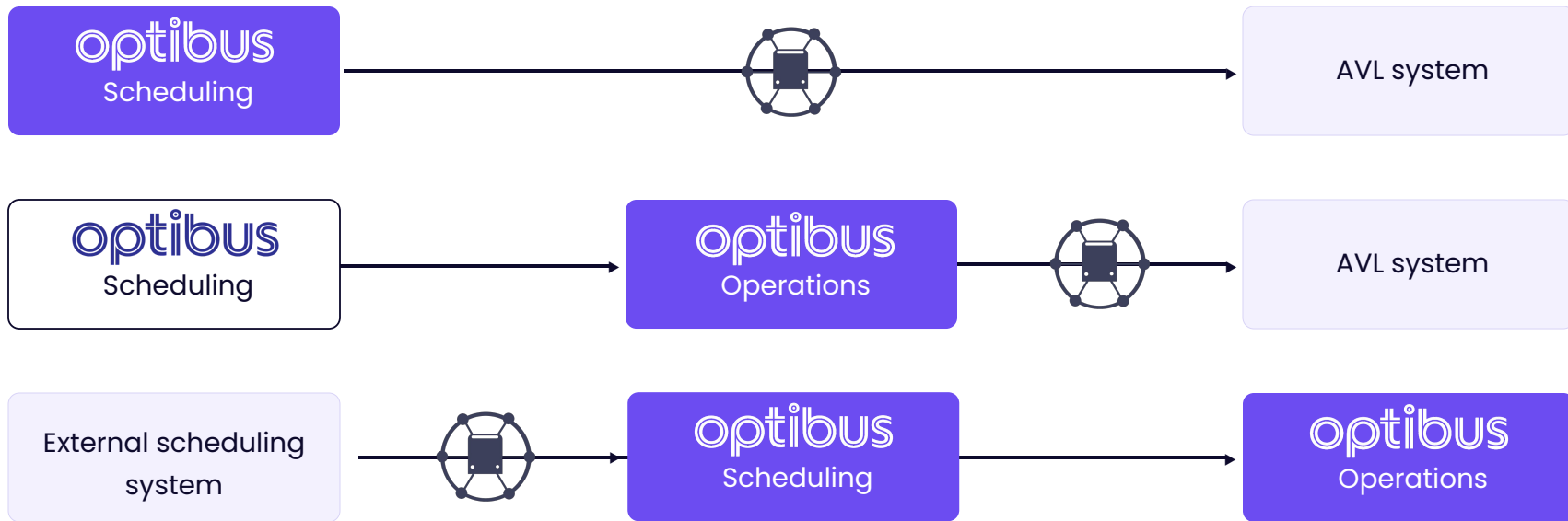
Import operational data into Optibus scheduling



- Work with existing schedules in Optibus Ops
- *Work-in-progress*: Support for rosters

Summary

TODS @ Optibus





Thank you

Timon Kelter, R&D Team Lead, Optibus

timon.kelter@optibus.com

tods-transit.org





Anne-Sophie Lachapelle

Master's researcher

Polytechnique Montréal

Using GTFS and TODS data to
estimate the effect of public transit service
characteristics on energy consumption



Using GTFS and TODS data to estimate the effect of service features on energy consumption

Anne-Sophie Lachapelle
Polytechnique Montréal
july 2026

The Electrification Challenge

- The science of predicting vehicle autonomy is off to a good start
 - Range management is still complex
 - High operational variability creates risk

- Using the right data at the right moment:
 - When are the decision made ?
 - What do we know then ?



Facilitate the integration of e-buses by leveraging operational data available at the vehicle assignment stage

SO1

- Identify service features of energy consumption available in operational data

SO2

- Aggregate features at block-level and service-level to assess energy risk

SO3

- Use refuelling cycles to perform a sensitivity analysis of block features

APPROACH

Four-Step Methodology

1.

Data extraction and enrichment

- Merging GTFS/TODS
- Inputs shapes points and stops elevation through API[1]

2.

Vehicle Task and service Reconstruction

Reconstruct complete service sequences and computing metrics

3.

Refueling cycles Analysis

Map diesel refueling data and operational data.

4.

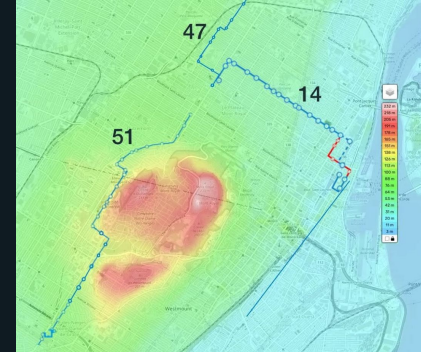
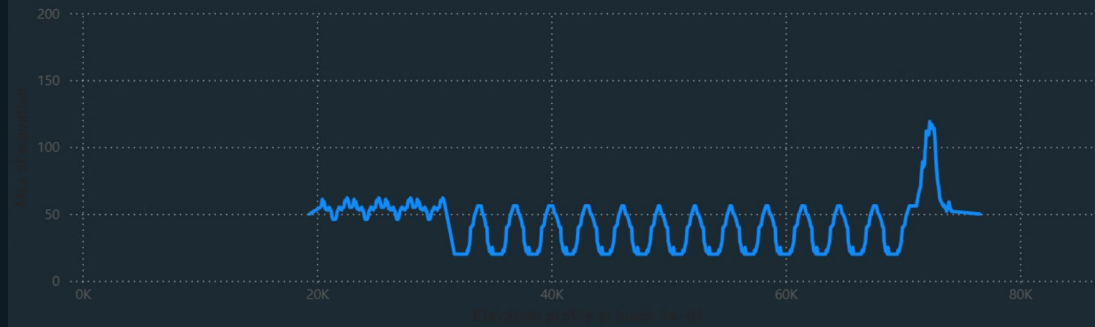
Risk Modelling

Classification algorithm assigns daily relative risk scores at vehicle assignment stage — probability an electric bus cannot complete the trip.

APPROACH

2. Vehicle Task and service Reconstruction

Elevation profile - Block 14-01



No impact
if canceled out



Positive impact



Negative impact



3. Refueling cycles Analysis



Research Impact

PRACTICAL BENEFITS

- Could allow agencies to analyze the extent the network or planning adjustments required for electrification
- Decision-support tool for depot staff to maximizes electric bus utilization while minimizing service quality risks

FURTHER ANALYSIS

- Risk on service level or service quality based on indicators extracted from GTFS+TODS
 - Passenger counting
 - Deadhead ratio
 - Headway ratio
 - etc.
- Service recovery capabilities following a range incident
- To be determined...

What you heard today



Josh

a large agency's honest
adoption journey



Jeff

the operator's case, and why
v2 works for railroads



Timon

a vendor already moving TODS
in and out of real products



Anne-Sophie

research the standard makes
possible, on real agency data

The thread: **the vendor side is ready; the missing ingredient is agencies asking**

Put TODS in your next scheduling or CAD/AVL procurement

SCHEDULING SYSTEM REQUIREMENTS

The scheduling system shall receive, store, provide access to, and make available via an appropriate API, data in the most recent version of the following open standards including their best practices: Rider-facing schedule data in **GTFS Schedule**; and As-operated data in the **Transit Operational Data Standard (TODS)**.

CAD/AVL SYSTEM REQUIREMENTS

The CAD/AVL system shall import, store and export access to the following data in the most recent version of the following open standards: Schedule data in the **Transit Operational Data Standard (TODS)** or GTFS-Schedule; As-operated data in the TIDES data suite; Passenger data in the TIDES data suite or GTFS-Ride; Real-time data on system performance in GTFS-Realtime.

MDIP Interoperable Procurement Resource · interoperablemobility.org/procurement

You are not asking vendors to invent something; **several already produce TODS on request**

Where to go from here

`tods-transit.org`

the spec, governance, resources

`github.com/MobilityData/
transit-operational-data-standard`

the specification on GitHub

`tods@mobilitydata.org`

MobilityData Slack · email the team

`tides-transit.org`

sibling standard TIDES, the historical record · June 2
TIDES Briefing recording

This recording will be shared with all registrants and posted publicly